# Indra Sanjaya

Fullstack Engineer | Environmental & Sustainability Engineer
Jakarta, Indonesia | +6282157358908 | indrasanjaya.ind@gmail.com | LinkedIn | GitHub | Portfolio

## SUMMARY

Engineer with two strong sides that work together. As a Fullstack Engineer, builds scalable web, mobile and AI-powered applications (RAG, LLM integration) with TypeScript, React, Next.js, Go and PostgreSQL for telecommunications systems. As an environmental, HSE and sustainability engineer with 8+ years in oil & gas and industrial sectors, brings expertise in regulatory compliance, risk management, GHG accounting, CDP climate disclosure and emission reduction strategy.

## WORK EXPERIENCE

**PT Technova Integrasi Solusi** Jakarta, Indonesia
*Fullstack Engineer* *September 2026 – Present*

- Developing end-to-end features for telecommunications projects using TypeScript, React, and Next.js.
- Building scalable backend services and REST APIs in Go and Node.js for telecom operational systems.
- Designing PostgreSQL schemas and optimizing queries to handle high-volume telecom data efficiently.

**PT Sepuluh Saudara Sejahtera** Kalimantan Timur, Indonesia
*Contract Engineer* *August 2023 – Present*

- Negotiating and managing suppliers for cost-effective, compliant procurement.
- Overseeing inventory planning and purchasing schedules to minimize waste and costs.
- Aligning procurement cross-functionally while optimizing vendor strategies.

**PT Pertamina Hulu Mahakam** Kalimantan Timur, Indonesia
*Sustainability Engineer* *August 2022 – August 2023*

- Led ESG compliance and ISO 14001 EMS, achieving Green and GOLD PROPER status.
- Coordinated conservation programs and environmental risk assessments per regulations.
- Managed environmental reporting to ensure full regulatory compliance.

**PT Pupuk Kalimantan Timur** Kalimantan Timur, Indonesia
*Environmental Engineer* *January 2019 – August 2022*

- Member of the Carbon Disclosure Project (CDP) team: identified emission sources, collected activity data across departments, and calculated Scope 1, 2 and 3 GHG emissions.
- Assessed climate-related risks and opportunities, set emission reduction targets and strategy, completed and submitted the CDP questionnaire, coordinated emission data verification, and reported to management and regulators.
- Reduced 700,000 tons $CO_2e$ (2020) through emissions control and energy efficiency.
- Led CEMS installation and monitored wastewater systems for regulatory compliance.
- Supported National GOLD PROPER, Green Industry Award (2020) and Sustainability Reporting.

**Beyond Petroleum (via PT Kontrol Power Utama & PT Wijaya Karya)** Papua, Indonesia
*Environmental Engineer* *August 2015 – December 2018*

- Conducted HSE inspections and supervised controls for high-risk site activities.
- Coordinated risk assessments and hazard controls across infrastructure projects.
- Upheld safety and environmental standards, achieving Zero LTI on the BP LNG Tangguh Tank Project.

## PROJECTS

**FireWork Studio - Social Media Manager** April 2026
AI-powered platform to track account and post performance, schedule content and discover trends, with a RAG-based AI storyboard system for planning and generating content ideas.
*Tech stack: Next.js, Prisma, REST API, PostgreSQL, Supabase, BetterAuth, Redis, Vercel, Cloudinary, Gemini API*

**SprintZee - Sportswear Marketplace** April 2026
Type-safe, scalable web app with MongoDB, JWT security and dynamic infinite scrolling.
*Tech stack: Next.js (App Router), React, TypeScript, MongoDB, Tailwind CSS, Zod, JWT, bcryptjs*

**TripPlanner - All-in-One Travel Organizer** April 2026
AI-powered web app for planning itineraries, securing data and generating intelligent suggestions.
*Tech stack: React.js, Vite, Tailwind CSS, Redux Toolkit, Node.js, Express.js, PostgreSQL, Sequelize, Google Generative AI, Jest*

**Voca - Real-Time Social Media Platform (via Expo Go)** April 2026
Scalable mobile social app for real-time content sharing, secure authentication and smooth navigation.
*Tech stack: React Native, Expo, Apollo Client, GraphQL, React Navigation, Expo Secure Store, EAS Build*

## SKILLS

**Language:** JavaScript, TypeScript, Go (Golang)

**Front End & Mobile:** React.js, Next.js (App Router), Redux Toolkit, Tailwind CSS, HTML, CSS, Apollo Client, React Native, Expo, React Navigation

**Back End & APIs:** Node.js, Express.js, REST API, GraphQL (Apollo Server), Swagger/OpenAPI

**Databases & ORM:** PostgreSQL, MongoDB, Redis, Prisma, Sequelize, Supabase

**Authentication & Security:** JWT, bcryptjs, Express Session, BetterAuth, Zod validation

**AI & LLM Integration:** Gemini API, OpenAI API, Vercel AI SDK, LangChain, LangGraph, RAG Systems (MongoDB Atlas Vector Search), Prompt Engineering, Semantic Search, AI Chat Interfaces

**Tools & Others:** Git, GitHub, Docker, Postman, Jest, Vite, Vercel, Firebase, Cloudinary, FullCalendar, Microsoft Office, AutoCAD, SAP

**Environmental & Sustainability:** ISO 14001 EMS, PROPER Compliance (Green & Gold), GHG Inventory & Carbon Accounting (Scope 1, 2, 3), CDP Climate Disclosure, Climate Risk & Opportunity Assessment, Emission Reduction Strategy, CEMS & Emissions Monitoring, Wastewater Management, Energy Efficiency, ESG & Sustainability Reporting, Environmental Risk Assessment

**HSE & Risk Management:** HSE Inspection & Auditing, Hazard Identification & Risk Assessment, High-Risk Work Supervision, Incident Prevention (Zero LTI), Contractor & Site Safety Coordination, Regulatory Compliance & Reporting

**Operations & Procurement:** Supplier Negotiation & Contract Management, MRP (Material Requirement Planning) & Inventory Optimization, Purchasing & Cost Analysis, Cross-functional Coordination

## CERTIFICATIONS

**HackerRank, May 2026:** Software Engineer | Frontend Developer (React) | SQL (Advanced) | REST API (Intermediate) | Node.js (Intermediate) | JavaScript (Intermediate) | Problem Solving (Intermediate) | SQL (Intermediate) | SQL (Basic)
**HackerRank, March 2026:** Software Engineer Intern | Problem Solving (Basic) | JavaScript (Basic) | React (Basic) | CSS (Basic)

## TECHNICAL WRITING

Articles at indrasanjaya.me/articles, in English and Indonesian: Circuit Breaker Pattern, RAG chatbot with MongoDB Atlas and Gemini, LangGraph CLI debugging on Windows, Metadata 101.

## EDUCATION

**Sea Labs Indonesia** — Jakarta, Indonesia
*Software Engineer Trainee – Labs Bootcamp Program (SWE Track)* — *July 2026 – Present*

**Hacktiv8** — Jakarta, Indonesia
*Full Stack JavaScript Immersive Program (Score 94/100 - Transcript)* — *November 2025 – April 2026*

**Universitas Mulawarman** — Samarinda, Indonesia
*Bachelor of Environmental Engineering (GPA 3.86/4.00 - Transcript)* — *2010 – 2014*